Mult-area OSPF

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Purpose:

This lab shows how to configure and test Multi-area OSPF understand, provides understanding on how Multi-area OSPF operates, which is like real-world enterprise and service providers networks

Background:

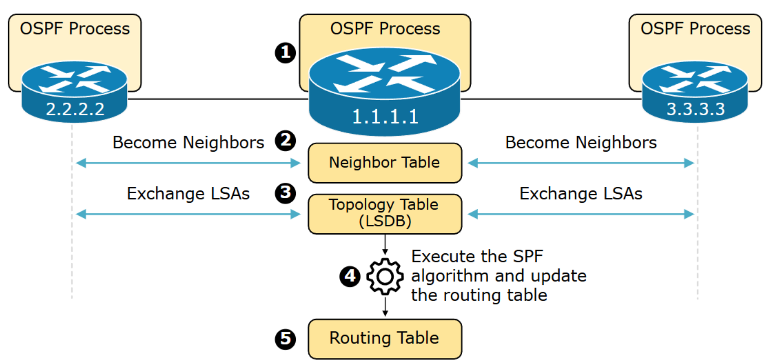
OSPF:

Before exploring multi-area OSPF, it is important to first understand core principles of OSPF; OPSF (Open Shortest Path First) is an open standard protocol, meaning pretty much any router will support it. It is an interior gateway protocol (IGP), meaning it is used within a single autonomous system (AS).

OSPF is also a link-state routing protocol, the goal of the a routing protocol is to learn routes, the way OSPF learns routes is by getting information about every router and subnet within the entire network , resulting in every router having the same information about the network as each other, the way routers get this information is by sending something called a link-state advertisement or a LSA, these LSAs contain information about the subnet router, and other network information. All information learned from the routers sending LSAs will be kept by OSPF in a link-state database, also known as a LSDB, the goal is to have each router contain the same information in their LSDBs.

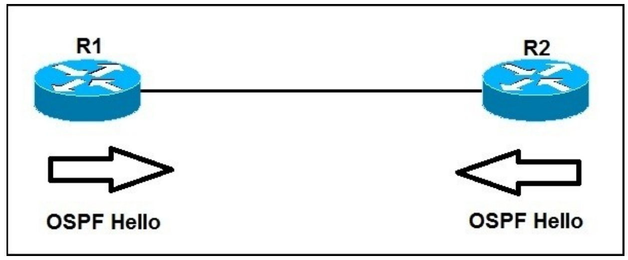
How it works:

1. Two routers running OSPF on same link agree to form a neighbor relationship
2. The neighbor routers will exchange their LSDB information with each other
3. Once the LSDB information has been exchanged, each router will choose the best route to add to its own routing table based on the LSDB information, done using a calculation called SPF



How connecting routers form neighbor relationships:

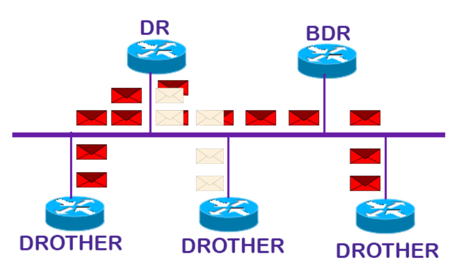
Before a relationship is formed each router needs a Router ID (RID), the RID is a number that can be used to identify an individual router, kind of like an OSPF name, it is in the format of an IPv4 address but can be set to whatever the user desires. A RID can be set manually or the router will assign one itself; a router will first check if a Router ID is already manually set, if not the router will choose the highest loopback address, if there are no loopback interfaces on the router, it will choose the highest non-loopback address.



Once OSPF is configured and each router has its own router IDs, routers in the network will start to reach out to potential neighbors by sending Hello messages, these are an essential part of the protocol's neighboring establishment process, with the message including crucial information like the unique router ID and any already known neighbors, the Hello messages also notifies the neighboring router of the current router still being up.

When another router receives the Hello Messages, it runs checks going through strict requirements before forming a neighboring relationship, with a required matched area ID, the connecting link of the 2 routers must also be on the same subnet, the Hello and Dead timers must be the same (The Dead timer is how long a router will wait before the router assumes its neighbor is "dead", with the default Hello timer being at 10 seconds), a matching authentication if used, and the stub area flag of the router must also match, routers will also need unique router IDs. If all these checks are passed, the receiving router will move to an in-it state and the router which originally received the Hello message will send Hello message of its own, this message not only lists its own router ID but it will also lists the original sender of the Hello message as a known neighbor, so when the original sender of the message receives, it see itself as a known neighbor, moving the original sender to the 2-way state. The original sender will send yet another Hello message, listing the receiving router as a known neighbor, allowing both routers to go into the 2-way state.

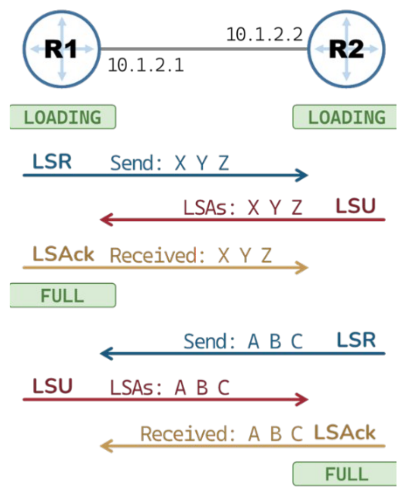
DR and BDRs:



DR and BDR are not elected on point-to-point connections, the election of a DR and BDR is based on a OSPF priority, by default it is set to 1, this value can be changed to influence the election of a DR and BDR. If the priorities of routers tie, the election is based on the highest router ID. When in the same segment, routers will only become full neighbors with DRs and BDRs, the other neighbors will remain in the 2-way state, meaning every router will ignore updates unless they come from a DR or BDR. This mechanism of having DR and BDRs prevents LSA storms, where too many LSAs can cause high CPU usage congestion and slow convergence, and potential routing failures.

Once the exchange process starts, routers will go into the Extart state, based on the router ID one router will control the sequence numbers and start the exchange process. After going into the exchange state, both neighboring routers will send each other a list of their LSAs, this is called a database description (DBD).

Next the routers will go into the loading state where the routers will investigate the DBD and request any information need it doesn't already have; this is done to prevent loops by having the routers having to request information instead of flooding the links with updates. The request the router makes after reviewing the DBD is called a link-state request (LSR), the other router will send back a link-state update (LSU). Finally, the router that sent the LSR / receives the LSR will send back a link-state acknowledgement (LSAck). This completes the process of exchanging LSAs, and the routers will go into a full neighborhood state.



Adding the best routes to the routing table:
The way OSPF determines the best route is by using a metric called cost, OSPF cost is given to a link based on the bandwidth of that interface.
Cost Calculation:

Refence bandwidth (Default of 100,000 Kbps) / Interface bandwidth

Common interface costs:
Serial: 1,544 Kbps, Cost of 64
Ethernet: 10,000 Kbps, Cost of 10
FastEthernet: 100,000 Kbps, Cost of 1

SPF Administrative Distances (ADs) Administrative Distance is a measure used by a router to select the best path when it learns the same route from two or more different routing protocols. It is essentially the trustworthiness rating of the routing information source, where lower is better.

Multi-Area OSPF:

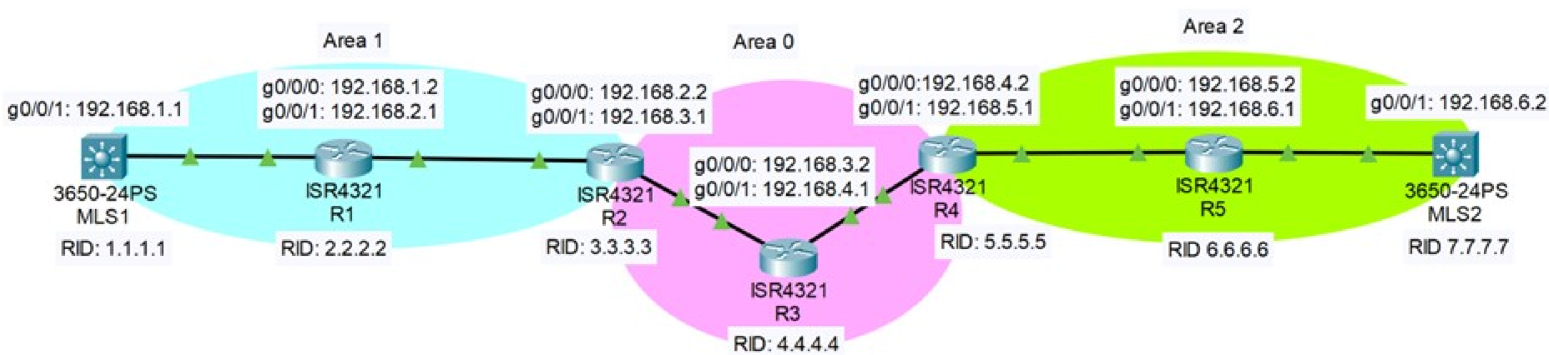
When using single area OSPF in a large many problems could occur, every router will have a copy of the same LSDB, filled with information about the entire network, most of it being unimportant to the specific router, this abundance of information can even fill up the disk space of the router. Not only will the LSDB be excessive, so will the routing table, with a route to each subnet every time a new packet comes in, a router might need to search through hundreds or even thousands of routes before it can find the right one, this also impacts the disk space and even hinders the processing power of the router itself. Another glaring issue with using single are OSPF in a large-scale network is whenever an update happens, it is flooded throughout the entire network, and each router will also have to rerun the SPF algorithm each time an update is sent. Thus, multi-area OSPF is crucial in larger networks.

Using multi-Area OSPF will reduce the sizes of each LSDB, summarize routing tables and will contain update messages to a single area. Each OSPF area is recommended to have less than 50 routers. Starting with Area 0, it is the backbone of the network, every other area must join the backbone area. Subnets should be unique to each OSPF area and not shared across multiple areas.

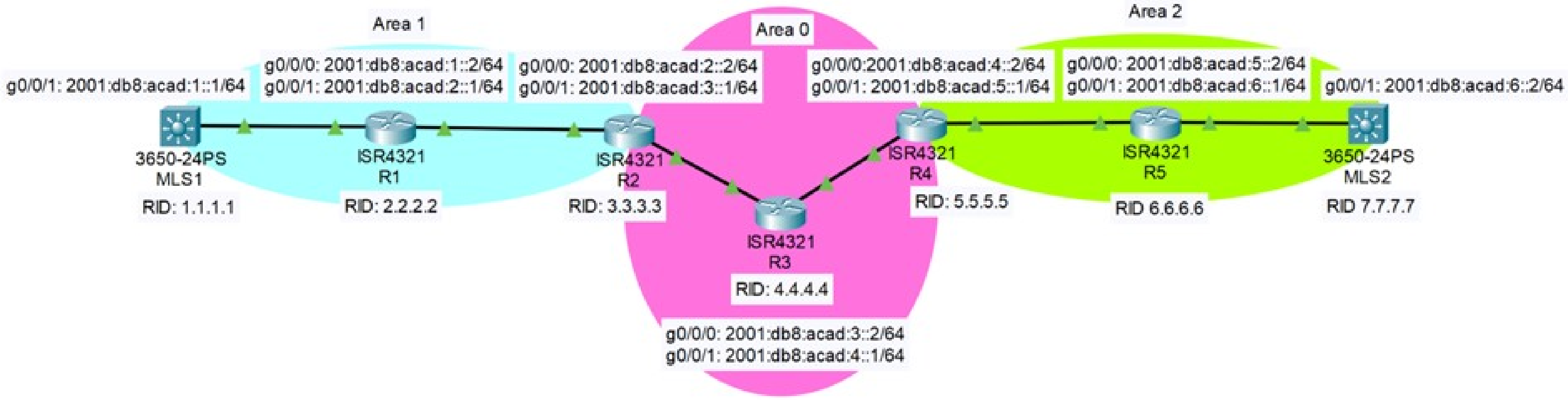
Routers with interfaces in 2 or more areas are called Area Border Routers (ABR), these routers are important since they are used to summarize routes from each area, reducing LSA floods, and separates LSDB per area, maintaining a unique link-state database per area. ABRs send a single area route to area 0, the backbone, the ABR will also summarize area 0’s subnet to other areas, this will also contain updates within its area, preventing a small change forcing every single router in the network to run the SPF algorithm again. Multi-Area OSPF is important to use in large networks to prevent overflooding of LSAs, keeping the size of routing tables manageable and giving the network more scalability.

Configuration:

IPv4 Topology:



IPv6 Topology:



The IPv4 and IPv6 topology both use 2 multilayer switches and 5 routers with multilayer switch 1, router 1 and router 2 being in area 1. The backbone/area 0 has router 2, router 3 and router 4. Router 4, Router 5 and multilayer switch 2 making up Area 2.

Different subnets in the topology:

Area 1:

Interface g0/0/1 of MLS1 and interface g0/0/0 of R1 are in the 192.168.1.0 / 2001:db8:acad:1::0 subnet. Interface g0/0/1 of R1 and interface g0/0/0 of R2 are in the 192.168.2.0 / 2001:db8:acad:2::0 subnet

Area 0:

Interface g0/0/1 of R2 and interface g0/0/0 of R3 are in the 192.168.3.0 / 2001:db8:acad:3::0 subnet. Interface g0/0/1 of R3 and interface g0/0/0 of R4 are in the 192.168.4.0 / 2001:db8:acad:4::0 subnet

Area 2:

Interface g0/0/1 of R4 and interface g0/0/0 of R5 are in the 192.168.5.0 / 2001:db8:acad:5::0 subnet. Interface g0/0/1 of R5 and int g0/0/1 of MLS2 are in the 192.168.6.0 / 2001:db8:acad:6::0 subnet

Configuration for routers:

Step 1: Assign IP addresses to interfaces:

```
Router(config)# interface GigabitEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
```

Step 2: Enable OSPF routing process:

```
Router(config)# router ospf 1
```

Step 3: Configure OSPF network statements by area:
Use the appropriate network and wildcard mask with area number. (A subnet mask of 255.255.255.0 would have a wildcard mask of 0.0.0.255)
The first network is in Area 0 (Backbone).
The second is in Area 1 (Non-backbone).
This configuration shows a router becoming an Area Border Router (ABR) since it connects two areas.

```
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router(config-router)# network 192.168.2.0 0.0.0.255 area 1
```

Step 4: (Optional to manually configure) Configure Router IDs

```
Router(config-router)# router-id 1.1.1.1
```

Step 5: Assign IP addresses to other interfaces (for Area 1)
An example of if this router is also connects to another network in Area 1:

```
Router(config)# interface GigabitEthernet0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
```

Step 6: (Optional) Use passive-interface to improve security/performance
If you don’t want OSPF hello packets sent out of interfaces that are not connected to other OSPF routers (e.g., user-facing interfaces):

```
Router(config-router)# passive-interface GigabitEthernet0/2
```

Alternatively you can assign the areas of each interface without using the network command:

```
Router(config)# router ospf 1
Router(config-router)# router-id 1.1.1.1

Router(config)# interface GigabitEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# ip ospf 1 area 0
Router(config-if)# no shutdown
```

```
Router(config)# interface GigabitEthernet0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# ip ospf 1 area 1
Router(config-if)# no shutdown
```

Configurations for multi-layer switches:

Step 1: Enable routing:

```
Switch(config)# ip routing
```

Step 2: Assign IP addresses to routed interfaces:

```
Switch(config)# interface GigabitEthernet0/1
Switch(config-if)# no switchport
Switch(config-if)# ip address 192.168.3.1 255.255.255.0
Switch(config-if)# no shutdown
```

Step 3: Enable OSPF:

```
Switch(config)# router ospf 1
Switch(config-router)# router-id 2.2.2.2
Switch(config-router)# network 192.168.3.0 0.0.0.255 area 0
```

Configuration files of routers 1-5:

Router 1:

```
Current configuration : 4197 bytes

! Last configuration change at 18:41:46 UTC Wed Sep 24 2025

version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core

hostname R1

boot-start-marker
boot-end-marker


vrf definition Mgmt-intf
 !
```

```
 address-family ipv4
 exit-address-family
 !
 address-family ipv6
 exit-address-family


no aaa new-model

login on-success log

subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated

crypto pki trustpoint TP-self-signed-702538597
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-702538597
 revocation-check none
 rsakeypair TP-self-signed-702538597

crypto pki trustpoint TP-self-signed-414644419
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-414644419
 revocation-check none
 rsakeypair TP-self-signed-414644419

crypto pki certificate chain TP-self-signed-702538597
 certificate self-signed 01
  3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
  30312E30 2C060355 04031325 494F532D 53656C66 2D536967 6E65642D
43657274
  69666963 6174652D 37303235 33383539 37301E17 0D323530 39323431
37353435
  325A170D 33303031 30313030 30303030 5A303031 2E302C06 03550403
1325494F
  532D5365 6C662D53 69676E65 642D4365 72746966 69636174 652D3730
32353338
  35393730 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
82010A02
  82010100 BED30968 E68F0ABA 32449B47 23325036 EBFEEE35 D596F9A9
C3CA2A58
  88D171D1 04018D2E 82AFF420 F2CB4826 40CC96B1 15892237 15FF2D0E
AE1E459A
```

```
  53BA4FC4 88636C29 86B98E02 E253ABB4 ABBBEEAF FBBB3B6E A2637EA3
9D1D9F71
     180548A6 8D16E11C 6F845234 1273CBA0 C7CD6A23 B2275E4A
F43B8B60 4F3DB08F
  AA378412 E66E9DEB E15293E6 221463A7 C13E5B64 9CC414C4 5BA31420
885D9AFB
  A6FE0093 E6BD21EA 0AD147E1 1A50FA65 ED28B94D CC175ACB B08C6F7B
CB9ED6B9
  973FEBA1 6C8DE8B9 AC6486D0 EE00F567 560E2E86 CEEBD59E CEE5F2D1
90C440FE
  DF11ACFF 4C9F26FB 72D156E9 4C6FE748 A063C429 4AD49436 E4B84979
F35B6791
  0C3A4F23 02030100 01A35330 51300F06 03551D13 0101FF04 05300301
01FF301F
  0603551D 23041830 16801490 B09D7018 8BA5705F 847CF9D2 DED3F104
19C4BF30
  1D060355 1D0E0416 041490B0 9D70188B A5705F84 7CF9D2DE D3F10419
C4BF300D
  06092A86 4886F70D 01010505 00038201 01007481 0274959D C6E0CD2A
C6C5318F
  84BF63DA CE5EF375 E704174F 59A50E7A 18B30510 9255F829 D77D6A15
7AE9BA2B
  20E0DE8F D813E2F7 0164CDC8 BDA93A5A 758787FB 0281AB70 328BB513
6B500F23
  4BD346E9 B9ABA994 721179AB 84C90715 A6C04C7A 7277AE18 5B286472
F32A9A96
  CBDB81E6 4D442537 63932348 D5391C5C 1AF6B14B 72AC09E2 E7B0C32C
05DB6F92
  D9A08CCE B2E49DD5 0BA4E2AA E3A7C307 0E923510 8D21CC29 D434C055
864590A1
  76DF5ADD 03ACC1E3 C38D3688 FD2AC0F5 FE5018CE 901C999C E6884F3D
7154BEB6
  C1BB9B53 BC45BF02 F87A3E58 393A0A79 64725550 91DB4B47 C94A4077
80067A5E
  E021D079 65CB6FEE F7C7A1C3 B4B27215 42B9
        quit
crypto pki certificate chain TP-self-signed-414644419

license udi pid ISR4321/K9 sn FDO21491LXF
no license smart enable
diagnostic bootup level minimal

spanning-tree extend system-id

redundancy
 mode none
```

```
interface GigabitEthernet0/0/0
 ip address 192.168.1.2 255.255.255.0
 negotiation auto
 ipv6 address 2001:DB8:ACAD:1::2/64
 ipv6 ospf 1 area 1

interface GigabitEthernet0/0/1
 ip address 192.168.2.1 255.255.255.0
 negotiation auto
 ipv6 address FE80::2 link-local
 ipv6 address 2001:DB8:ACAD:2::1/64
 ipv6 ospf 1 area 1

interface Serial0/1/0
 no ip address
 shutdown

interface Serial0/1/1
 no ip address
 shutdown

interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 shutdown
 negotiation auto

router ospfv3 1
 router-id 2.2.2.2

 address-family ipv6 unicast
 exit-address-family

router ospf 1
 router-id 2.2.2.2
 network 192.168.1.0 0.0.0.255 area 1
 network 192.168.2.0 0.0.0.255 area 1

ip forward-protocol nd
ip http server
```

```
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0

control-plane

line con 0
 transport input none
 stopbits 1
line aux 0
 stopbits 1
line vty 0 4
 login

end
```

Router 2:

```
Building configuration...

Current configuration : 1998 bytes

 Last configuration change at 17:40:40 UTC Wed Sep 24 2025

version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
no platform punt-keepalive disable-kernel-core

hostname R2

boot-start-marker
boot system bootflash:isr4300-universalk9.16.09.08.SPA.bin
boot system flash isr4300-universalk9.16.09.08.SPA.bin
boot-end-marker

vrf definition Mgmt-intf

 address-family ipv4
 exit-address-family

 address-family ipv6
 exit-address-family


no aaa new-model
```

```
no ip domain lookup

login on-success log

subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated

license udi pid ISR4321/K9 sn FLM240607Q1
no license smart enable
diagnostic bootup level minimal

spanning-tree extend system-id

redundancy
 mode none

interface GigabitEthernet0/0/0
 ip address 192.168.2.2 255.255.255.0
 ip ospf 1 area 1
 negotiation auto
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:2::2/64
 ipv6 ospf 1 area 1

interface GigabitEthernet0/0/1
 ip address 192.168.3.1 255.255.255.0
 ip ospf 1 area 0
 negotiation auto
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:3::1/64
 ipv6 ospf 1 area 0

interface GigabitEthernet0/1/0
 no ip address
 shutdown
 negotiation auto

interface GigabitEthernet0/1/1
 no ip address
 shutdown
 negotiation auto

interface GigabitEthernet0
```

```
 vrf forwarding Mgmt-intf
 no ip address
 shutdown
 negotiation auto

router ospfv3 1
 router-id 3.3.3.3

 address-family ipv6 unicast
 exit-address-family

router ospf 1
 router-id 3.3.3.3
 network 192.168.2.0 0.0.0.255 area 1
 network 192.168.3.0 0.0.0.255 area 0

ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0

control-plane

line con 0
 transport input none
 stopbits 1
line aux 0
 stopbits 1
line vty 0 4
 login

End
```

Router 3:

```
Building configuration...


Current configuration : 4069 bytes

Last configuration change at 17:47:14 UTC Tue Sep 23 2025

version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
no platform punt-keepalive disable-kernel-core
```

```
hostname R3

boot-start-marker
boot-end-marker

vrf definition Mgmt-intf

 address-family ipv4
 exit-address-family

 address-family ipv6
 exit-address-family

no logging console

no aaa new-model

no ip domain lookup

login on-success log

subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated

crypto pki trustpoint TP-self-signed-2667303412
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-2667303412
 revocation-check none
 rsakeypair TP-self-signed-2667303412

crypto pki certificate chain TP-self-signed-2667303412
 certificate self-signed 01
  30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
  31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
  69666963 6174652D 32363637 33303334 3132301E 170D3235 30393232
31353134
  34365A17 0D333030 31303130 30303030 305A3031 312F302D 06035504
03132649
  4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32
36363733
```

```
  30333431 32308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
  0A028201 0100C00D 97512689 5C5C0F2C 5B219341 524A538C E23ADF5A
79133BC6
  072DDA3E CCB1673E 340B694F 2A900E41 4ABBD58B 3481CEC8 E4723EA2
1EB45A14
  5261BDE6 00E8557C C10D03BA 0B355A62 5342F140 507B74E8 D49E49D9
3BEC9CAE
  14F931B6 762FFB38 1A5CB094 F98456CF 748100F4 225B2529 777247F9
486701AF
  17AA5C3C F70AC43A C41A8BAD 96FD8AE3 5B30F3AB D82D672B 62143662
88473F00
  2B6E64B1 52458A06 A151E183 4B2B0BF4 864BB132 704BCFB6 B562BA93
2D1640FC
  609AF66A 83602E10 6DBC6526 44FB50BF 4138EB18 F28F0111 8923823D
53D4CE86
  178208E7 B1DCA1ED 20939FBD 32F6A1EE AB737B65 B1AD77DE 262C93E5
2C89B456
  2A0C6054 C34F0203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
  301F0603 551D2304 18301680 1412175C E34A17D4 A94B1933 5B5571E9
DA037A43
  06301D06 03551D0E 04160414 12175CE3 4A17D4A9 4B19335B 5571E9DA
037A4306
  300D0609 2A864886 F70D0101 05050003 82010100 34C1E23A CB48280D
3DBD92BC
  84EC97C7 95601B0D 637F8533 55AC4348 A51F5EEB C084D687 20E41A44
7F5DDEB1
  529F76B1 92F016CA 7C8A72ED 526D0F7E C7306991 535301AF B37D7F4F
A48B709B
  A4BA9390 E4699A09 DCF32595 EA374B01 00B1670E BE90DE03 49CB2FD6
A42C8318
  9C9DCA67 C0A9E568 4266C04F 3D9AD12F 1E92E14D E505A05B 4792E6B1
899CE4C6
  2B60491B 84B8A95F 344E0A52 F7ED6430 116618B9 492177FE 84B49F5D
22E8A6EB
  AADC4BA7 BB91EF7D 91CDE9FB C9EA4975 71D5EA8C 31AC01F0 186DC6B0
73B8FA7A
  2BA63BB3 41541B3A A2275861 65EB9529 ADB78013 1368A719 FD59D7F4
2B9F8016
  9BD5F334 8AEC42D2 42C6F30F EF32F782 14A13732
        quit

license udi pid ISR4321/K9 sn FLM2407011F
no license smart enable
diagnostic bootup level minimal
```

```
spanning-tree extend system-id

redundancy
 mode none

interface GigabitEthernet0/0/0
 ip address 192.168.3.2 255.255.255.0
 ip ospf 1 area 0
 negotiation auto
 ipv6 address 2001:DB8:ACAD:3::2/64
 ipv6 ospf 1 area 0

interface GigabitEthernet0/0/1
 ip address 192.168.4.1 255.255.255.0
 ip ospf 1 area 0
 negotiation auto
 ipv6 address 2001:DB8:ACAD:4::1/64
 ipv6 ospf 1 area 0

interface GigabitEthernet0/1/0
 no ip address
 shutdown
 negotiation auto

interface GigabitEthernet0/1/1
 no ip address
 shutdown
 negotiation auto

interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 shutdown
 negotiation auto

router ospfv3 1
 router-id 4.4.4.4

 address-family ipv6 unicast
 exit-address-family

router ospf 1
 router-id 4.4.4.4
 network 192.168.3.0 0.0.0.255 area 0
 network 192.168.4.0 0.0.0.255 area 0

ip forward-protocol nd
```

```
no ip http server
ip http authentication local
no ip http secure-server
ip tftp source-interface GigabitEthernet0

control-plane

line con 0
 transport input none
 stopbits 1
line aux 0
 stopbits 1
line vty 0 4
 login

end
```

Router 4:

```
Building configuration...

Current configuration : 4305 bytes

! Last configuration change at 18:10:23 UTC Wed Sep 24 2025

version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core

hostname R4

boot-start-marker
boot-end-marker

vrf definition Mgmt-intf

 address-family ipv4
 exit-address-family

 address-family ipv6
 exit-address-family

no aaa new-model

login on-success log
```

```
subscriber templating

vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated

crypto pki trustpoint TP-self-signed-2949602955
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-2949602955
 revocation-check none
 rsakeypair TP-self-signed-2949602955

crypto pki trustpoint TP-self-signed-1275915594
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-1275915594
 revocation-check none
 rsakeypair TP-self-signed-1275915594

crypto pki certificate chain TP-self-signed-2949602955
 certificate self-signed 01
  30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
  31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
  69666963 6174652D 32393439 36303239 3535301E 170D3235 30393232
31353133
  35345A17 0D333030 31303130 30303030 305A3031 312F302D 06035504
03132649
  4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32
39343936
  30323935 35308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
  0A028201 0100B4C7 DA1358C4 A3B6E899 C18F07C0 184D3B46 E8F17624
149A5FED
  EE662E04 5C8A2D38 DD0EF867 536B9EC1 23ACEDBD FA386BB7 9E7A73C2
C5485CE9
  9D453DEE 5F81C6F5 56803D9A 1C7C6464 BDBEC081 70CF2167 70D8E939
9DDDBE5F
  0E622D70 EDC34B88 0857AA65 0A2A9117 684F26AB 8E6C54BE 6D9A6D88
C2102588
  F6D98E6C 87E61DCC 1ACCFCFA 02875587 586A5FA5 7E05D2EF 0A2875E6
54FEF9E2
  7C927FB8 DB428D8B 35D43DB9 8C6868B0 F2C4BF3D 428C909C F901DBD1
9C564C72
```

```
  3E6C9012 3E7C36B0 7A732099 B6B700B4 AA8F1A66 BCE66657 EEC07639
57FA1266
  5BE32385 50A817BB F76C79AD AFBF087E CE7792BE 0D7584BC 68831EC9
E39961AD
  1B2B1595 F3DD0203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
  301F0603 551D2304 18301680 1417A0CD 5AC74A75 6E64D701 1240F5E4
54F48E3B
  C4301D06 03551D0E 04160414 17A0CD5A C74A756E 64D70112 40F5E454
F48E3BC4
  300D0609 2A864886 F70D0101 05050003 82010100 779E9A91 1064F2FE
2CDFE308
  30420E99 6A8D09C0 AA8A489E A1F04D20 6F0650DA 3AF3D99F 4C96E3CA
97275995
  EBC89F57 12D2CE58 24A901B1 4AD6212E 8FC7290A 5ABD8181 A4B9D04B
A77564E0
  1B12E33D 7AD3AD7C 8C75A752 77358D39 242CFC14 1C2CBBBD 4B437283
409271BB
  2E71CCDC 8F63A892 4BCAD3A8 5C54EF5B 3CF8F041 35D7A6AB 04038FCB
A2BCAE8D
  DAD41F8D A9B6A90F 3800D570 69628618 FBF568D0 6C344158 73A18678
CD241770
  F99898FB 17C4E9E0 3C17E771 5DCC8936 FDF0141D 9720B028 15DEE38A
9E56B029
  4704AC8B 6820A530 5E4B51EA BC72B196 07CC7431 CDE2AF57 33075329
7833AA59
  DBD3A610 77A74D4E 8AF043E1 99BBBB35 4CE10CDC
        quit
crypto pki certificate chain TP-self-signed-1275915594

license udi pid ISR4321/K9 sn FDO214420HW
license boot level appxk9
no license smart enable
diagnostic bootup level minimal

spanning-tree extend system-id




redundancy
 mode none

interface GigabitEthernet0/0/0
 ip address 192.168.4.2 255.255.255.0
 ip ospf 1 area 0
 negotiation auto
```

```
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:4::2/64
 ipv6 ospf 1 area 0

interface GigabitEthernet0/0/1
 ip address 192.168.5.1 255.255.255.0
 ip ospf 1 area 2
 negotiation auto
 ipv6 address 2001:DB8:ACAD:5::1/64
 ipv6 ospf 1 area 2

interface Serial0/1/0
 no ip address
 shutdown

interface Serial0/1/1
 no ip address
 shutdown

interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 shutdown
 negotiation auto

router ospfv3 1
 router-id 5.5.5.5

 address-family ipv6 unicast
 exit-address-family

router ospf 1
 router-id 5.5.5.5
 network 192.168.4.0 0.0.0.255 area 0
 network 192.168.5.0 0.0.0.255 area 2

ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0

control-plane

line con 0
 transport input none
 stopbits 1
```

```
line aux 0
 stopbits 1
line vty 0 4
 login

End
```

Router 5:

```
Building configuration...

Current configuration : 4274 bytes

! Last configuration change at 18:11:18 UTC Wed Sep 24 2025

version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
no platform punt-keepalive disable-kernel-core

hostname R5

boot-start-marker
boot system flash bootflash:isr4300-universalk9.16.09.08.SPA.bin
boot-end-marker

vrf definition Mgmt-intf

 address-family ipv4
 exit-address-family

 address-family ipv6
 exit-address-family

no aaa new-model

no ip domain lookup

ip dhcp pool webuidhcp

login on-success log

subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
```

```
multilink bundle-name authenticated

crypto pki trustpoint TP-self-signed-214699181
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-214699181
 revocation-check none
 rsakeypair TP-self-signed-214699181

crypto pki trustpoint TP-self-signed-2219300048
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-2219300048
 revocation-check none
 rsakeypair TP-self-signed-2219300048

crypto pki certificate chain TP-self-signed-214699181
 certificate self-signed 01
  3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
  30312E30 2C060355 04031325 494F532D 53656C66 2D536967 6E65642D
43657274
  69666963 6174652D 32313436 39393138 31301E17 0D323530 39323231
35323833
  395A170D 33303031 30313030 30303030 5A303031 2E302C06 03550403
1325494F
  532D5365 6C662D53 69676E65 642D4365 72746966 69636174 652D3231
34363939
  31383130 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
82010A02
  82010100 9C8CDF75 E62B61E4 2DA62E68 A41A3565 04EB0CB1 07173EE7
7C18324B
  D0B4B50A F5710915 F488CEB8 CD8D4C24 3F2D8E33 B000BA5B C0E80B57
64CD8A47
  2EE86E3C D3D1DD46 9451EB69 0047CC83 95DB52DC 58212549 8CDB1752
158B2659
  1F0CD474 1B9E9F50 0E0B35F8 D9E0B395 217B9FD4 4B69EC0F 01AE5142
FD30EA2C
  929F6120 A99A5102 FFCDE157 7E667705 51F9DA69 FBA56A34 57425A7F
B51EDC65
  A40903C3 96D72924 0A0B36B3 663B53C9 64595697 3D4AFB2D 6BBEB5B9
60DB060F
  B90B9526 7E226250 F4FE561B 3BAC7413 65F34724 216E2960 25E21778
6ADE1822
  C74BF50B B0D65C09 FE684EC5 3E22C0C3 1D594F5C C6C5340C 3824E672
9B39B57F
  C30533BD 02030100 01A35330 51300F06 03551D13 0101FF04 05300301
01FF301F
```

```
  0603551D 23041830 16801455 5D132A91 26FDDD4B E325A7B9 1461E5C2
60A20B30
  1D060355 1D0E0416 0414555D 132A9126 FDDD4BE3 25A7B914 61E5C260
A20B300D
  06092A86 4886F70D 01010505 00038201 01003136 E908611B 1D673D6E
E3DF97E3
  6643332E 8C3495BC FA73A34D D85BFD06 9BB820C1 CCA8DC2D 2177E3DA
AB29F306
  1F8363B6 C70A37DB B2625C0C 2BA6675A 6FAED4E4 9C685E0F 555183E6
B4C57D07
  7CC18A0B BD3B11BA 2695F9E7 BB0A7304 46D2B661 2107D936 9F5D0A87
ECBA8692
  D5827AF1 B2F5B9CF 5166E18F 1EEC73CE CE3E8C87 A2989D17 DE52A2F4
7FC40026
  27045347 598DF5CA 3752DC1A 9682DA7A 82F95901 809BAAA0 6BF0936F
4C147FEE
  689F26F3 B373BF11 024E1B8D 8E89C91F F6B0198E B1A89594 5AB2C4A5
341D5866
  678FEF02 8EE6941E 62297457 05563168 C45F303B 3DC4307C 929ECC1F
EF83A730
  86705230 D049A5D0 40BCBD35 C450B32B F1F5
        quit
crypto pki certificate chain TP-self-signed-2219300048
!
license udi pid ISR4321/K9 sn FDO21442167
no license smart enable
diagnostic bootup level minimal

spanning-tree extend system-id

redundancy
 mode none

interface GigabitEthernet0/0/0
 ip address 192.168.5.2 255.255.255.0
 ip ospf 1 area 2
 negotiation auto
 ipv6 address 2001:DB8:ACAD:5::2/64
 ipv6 ospf 1 area 2

interface GigabitEthernet0/0/1
 ip address 192.168.6.1 255.255.255.0
 ip ospf 1 area 2
 negotiation auto
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:6::1/64
 ipv6 ospf 1 area 2
```

```
interface Serial0/1/0
 no ip address
 shutdown

interface Serial0/1/1
 no ip address
 shutdown

interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 shutdown
 negotiation auto

router ospfv3 1
 router-id 6.6.6.6
 !
 address-family ipv6 unicast
 exit-address-family

router ospf 1
 router-id 6.6.6.6

ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0

control-plane

line con 0
 transport input none
 stopbits 1
line aux 0
 stopbits 1
line vty 0 4
 login

End
```

Configuration files of multi-layer switches 1 and 2:

Switch 1:

```
Building configuration...
```

```
Current configuration : 8936 bytes

! Last configuration change at 17:37:01 UTC Wed Sep 24 2025

version 16.9
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
! Call-home is enabled by Smart-Licensing.
service call-home
no platform punt-keepalive disable-kernel-core

hostname MLS1

vrf definition Mgmt-vrf

 address-family ipv4
 exit-address-family

 address-family ipv6
 exit-address-family

no aaa new-model
switch 1 provision c9200l-24t-4

ip routing
ip arp entry learn 10240

login on-success log
ipv6 unicast-routing
call-home
 ! If contact email address in call-home is configured as sch-
smart-licensing@cisco.com
 ! the email address configured in Cisco Smart License Portal
will be used as contact email address to send SCH notifications.
 contact-email-addr sch-smart-licensing@cisco.com
 profile "CiscoTAC-1"
  active
  destination transport-method http
  no destination transport-method email

crypto pki trustpoint TP-self-signed-11466316
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-11466316
 revocation-check none
 rsakeypair TP-self-signed-11466316
```

```
crypto pki trustpoint SLA-TrustPoint
 enrollment pkcs12
 revocation-check crl

crypto pki trustpoint TP-self-signed-3874827068
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-3874827068
 revocation-check none
 rsakeypair TP-self-signed-3874827068


crypto pki certificate chain TP-self-signed-11466316
 certificate self-signed 01
  3082032C 30820214 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
  2F312D30 2B060355 04031324 494F532D 53656C66 2D536967 6E65642D
43657274
  69666963 6174652D 31313436 36333136 301E170D 32353039 32333137
33363535
  5A170D33 30303130 31303030 3030305A 302F312D 302B0603 55040313
24494F53
  2D53656C 662D5369 676E6564 2D436572 74696669 63617465 2D313134
36363331
  36308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201
  0100B8DE BA3C009F C2ABAC00 EE041E8B 4F306F00 68D1EDB6 08FAC493
51B3385C
  DE7A6851 6711535E 36925785 5C484535 6214C202 EA8A0410 2999C6A1
25FF43D5
  36ACA908 BD3C19F6 F0415DC9 596A4F3A 542EBCBE 9F888237 4DB79BDE
7B628279
  68D0F21D EDD022D2 0B70079C A42D141E D8DF2E6E 5798F069 98F34F55
38BF62AF
  C5DBB3E5 2E14174E C9774158 2C4932D3 7E2617B0 A1FB5034 6732B968
25862EEB
  6B141B9A 91CA258B 270A8B03 8FD8B444 9390C93C 4AE234C5 7C1D4AA2
B3219392
  B2FF327D B4E0515E 136D7818 3EC9D770 44E5C066 D0F21DD5 FCA203A7
98C51AB1
  FA6ED9D9 BEA4E9A5 54D4ACA9 9E0A3C9F 8457F96F 9D728113 9D65FDD8
80576FFF
  51690203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603
  551D2304 18301680 14148D12 4284A4A5 A43C53D4 0F891806 B269B2D0
29301D06
  03551D0E 04160414 148D1242 84A4A5A4 3C53D40F 891806B2 69B2D029
300D0609
```

```
  2A864886 F70D0101 05050003 82010100 138D7588 66B97D2E 88C6F77A
4B186D48
  44425970 78AC160B 80A608F4 D02297BE 3B4E4E24 8335010E D4A4A54F
3BED49AA
  036CBED5 E7AACB10 9094E6DC 8E874DB3 C052C5ED 0745BE6C EE304ACD
794DFC2A
  D1BBF743 FE73203A 930B7447 F8817F70 3A4EBF9E DFA39FE1 8C9656AC
8100A1A2
  F07A45B9 92028F6F 1C45B048 215E1797 8905A97E 9A37FDE6 E132819A
EAAA178E
  AD9BB2B6 291F453E 649244CD 2989A38C A3BE6321 EEBE0586 E46E26FE
F3966E42
  6599576D D4A827EB A8926248 EADCAB41 43F70E58 59D95939 12DD3D51
762A126D
  EFCCD1A7 F20DDDFA D03EDFEE CCDDFF82 EE887F07 552BD2C9 BCF855FA
32AB82BC
  70A2EDE1 363AEB8D 77E1D0CB 475E45DC
        quit
crypto pki certificate chain SLA-TrustPoint
 certificate ca 01
  30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101
0B050030
  32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317
43697363
  6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533
30313934
  3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355
040A1305
  43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73
696E6720
  526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382
010F0030
  82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1
F1EFF64D
  CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388
8A38E520
  1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7
D8F256EE
  4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F
EA2956AC
  7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF
58BD7188
  68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B
42C68BB7
  C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8
8F27D191
```

```
  C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368
95135E44
  DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04
04030201
  06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604
1449DC85
  4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01
010B0500
  03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78
240DA905
  604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1
6C9E3D8B
  D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146
8DFC66A8
  467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2
55A9232C
  7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49
1765308B
  5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69
39F08678
  80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD
230E3AFB
  418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F
719BB2F0
  D697DF7F 28
        quit
crypto pki certificate chain TP-self-signed-3874827068

license boot level network-essentials addon dna-essentials

diagnostic bootup level minimal

spanning-tree mode rapid-pvst
spanning-tree extend system-id

redundancy
 mode sso

class-map match-any system-cpp-police-topology-control
  description Topology control
class-map match-any system-cpp-police-sw-forward
  description Sw forwarding, L2 LVX data, LOGGING
class-map match-any system-cpp-default
  description Inter FED, EWLC control, EWLC data
class-map match-any system-cpp-police-sys-data
  description Learning cache ovfl, High Rate App, Exception, EGR
Exception, NFL SAMPLED DATA, RPF Failed
```

```
class-map match-any system-cpp-police-punt-webauth
  description Punt Webauth
class-map match-any system-cpp-police-l2lvx-control
  description L2 LVX control packets
class-map match-any system-cpp-police-forus
  description Forus Address resolution and Forus traffic
class-map match-any system-cpp-police-multicast-end-station
  description MCAST END STATION
class-map match-any system-cpp-police-multicast
  description Transit Traffic and MCAST Data
class-map match-any system-cpp-police-l2-control
  description L2 control
class-map match-any system-cpp-police-dot1x-auth
  description DOT1X Auth
class-map match-any system-cpp-police-data
  description ICMP redirect, ICMP_GEN and BROADCAST
class-map match-any system-cpp-police-stackwise-virt-control
  description Stackwise Virtual
class-map match-any non-client-nrt-class
class-map match-any system-cpp-police-routing-control
  description Routing control and Low Latency
class-map match-any system-cpp-police-protocol-snooping
  description Protocol snooping
class-map match-any system-cpp-police-dhcp-snooping
  description DHCP snooping
class-map match-any system-cpp-police-system-critical
  description System Critical and Gold Pkt

policy-map system-cpp-policy

interface GigabitEthernet0/0
 vrf forwarding Mgmt-vrf
 no ip address
 shutdown
 negotiation auto

interface GigabitEthernet1/0/1
 no switchport
 ip address 192.168.1.1 255.255.255.0
 ipv6 address 2001:DB8:ACAD:1::1/64
 ipv6 ospf 1 area 1

interface GigabitEthernet1/0/2

interface GigabitEthernet1/0/3

interface GigabitEthernet1/0/4
```

```
interface GigabitEthernet1/0/5

interface GigabitEthernet1/0/6

interface GigabitEthernet1/0/7

interface GigabitEthernet1/0/8

interface GigabitEthernet1/0/9

interface GigabitEthernet1/0/10

interface GigabitEthernet1/0/11

interface GigabitEthernet1/0/12

interface GigabitEthernet1/0/13

interface GigabitEthernet1/0/14

interface GigabitEthernet1/0/15

interface GigabitEthernet1/0/16

interface GigabitEthernet1/0/17

interface GigabitEthernet1/0/18

interface GigabitEthernet1/0/19

interface GigabitEthernet1/0/20

interface GigabitEthernet1/0/21

interface GigabitEthernet1/0/22

interface GigabitEthernet1/0/23

interface GigabitEthernet1/0/24

interface GigabitEthernet1/1/1

interface GigabitEthernet1/1/2

interface GigabitEthernet1/1/3
```

```
interface GigabitEthernet1/1/4

interface Vlan1
 no ip address
 shutdown

router ospfv3 1
 router-id 1.1.1.1

 address-family ipv6 unicast
  router-id 1.1.1.1
 exit-address-family

router ospf 1
 router-id 1.1.1.1
 network 192.168.1.0 0.0.0.255 area 1

ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server

control-plane
 service-policy input system-cpp-policy

line con 0
 stopbits 1
line aux 0
 stopbits 1
line vty 0 4
 login
line vty 5 15
 login

End
```

Switch 2:

```
Building configuration...

Current configuration : 9538 bytes

! Last configuration change at 18:25:16 UTC Tue Sep 23 2025

version 16.9
no service pad
service timestamps debug datetime msec
```

```
service timestamps log datetime msec
! Call-home is enabled by Smart-Licensing.
service call-home
platform punt-keepalive disable-kernel-core

hostname MLS2

vrf definition Mgmt-vrf

 address-family ipv4
 exit-address-family

 address-family ipv6
 exit-address-family

no aaa new-model
switch 1 provision c9200l-24t-4g

ip routing
ip arp entry learn 10240

no ip domain lookup

login on-success log
ipv6 unicast-routing
call-home
 ! If contact email address in call-home is configured as sch-
smart-licensing@cisco.com
 ! the email address configured in Cisco Smart License Portal
will be used as contact email address to send SCH notifications.
 contact-email-addr sch-smart-licensing@cisco.com
 profile "CiscoTAC-1"
  active
  destination transport-method http
  no destination transport-method email

crypto pki trustpoint SLA-TrustPoint
 enrollment pkcs12
 revocation-check crl

crypto pki trustpoint TP-self-signed-226317576
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-226317576
 revocation-check none
 rsakeypair TP-self-signed-226317576

crypto pki trustpoint TP-self-signed-4227613475
```

```
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-4227613475
 revocation-check none
 rsakeypair TP-self-signed-4227613475

crypto pki certificate chain SLA-TrustPoint
 certificate ca 01
  30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101
0B050030
  32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317
43697363
 6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533
30313934
  3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355
040A1305
  43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73
696E6720
  526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382
010F0030
  82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1
F1EFF64D
  CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388
8A38E520
  1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7
D8F256EE
  4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F
EA2956AC
  7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF
58BD7188
  68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B
42C68BB7
  C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8
8F27D191
  C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368
95135E44
  DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04
04030201
  06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604
1449DC85
  4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01
010B0500
  03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78
240DA905
  604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1
6C9E3D8B
  D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146
8DFC66A8
```

```
  467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2
55A9232C
  7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49
1765308B
  5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69
39F08678
  80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD
230E3AFB
  418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F
719BB2F0
  D697DF7F 28
        quit
crypto pki certificate chain TP-self-signed-226317576
 certificate self-signed 01
  3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
  30312E30 2C060355 04031325 494F532D 53656C66 2D536967 6E65642D
43657274
  69666963 6174652D 32323633 31373537 36301E17 0D323530 39313931
37323034
  315A170D 33303031 30313030 30303030 5A303031 2E302C06 03550403
1325494F
  532D5365 6C662D53 69676E65 642D4365 72746966 69636174 652D3232
36333137
  35373630 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
82010A02
  82010100 B78C7504 76D79F4E EEECB917 F7E92CAF 953A9E5C 0CDEA82D
2B729012
  B04243A4 69030689 24AA8ACB C89DB360 F6A44E47 47E97CBC 20AE35A1
CAC6E6AD
  18612CE8 AB79C8BC 681F3D89 3A375616 3DCBF001 937F4B83 B734751F
0F6A3D5A
  29A4150E FD2690D2 870ECF60 8381FC8F 068330F6 97E42DD3 762B6A74
D887F4C7
  7B6B9A4C 51673575 C20C46BE 6A2B0CF5 0DC5217C 928A4F0C 30161C77
E0496A56
  41815AAE 22DD7CA9 86E1389D B7410677 02554D16 1B860FB7 45E4FA51
A750A793
  1EE6F791 15981B3C 6B58CE48 618325E4 CE6DC79A F32452BA C0D1CEA4
D5A67C89
  4CF9887A 9B2C9547 DED9D1DE 4C93C8A1 0A01FD7F 2338F052 D25A5294
16AD1EC3
  6058B4CF 02030100 01A35330 51300F06 03551D13 0101FF04 05300301
01FF301F
  0603551D 23041830 168014F3 C21951DC A12729C0 AA52857A DCC4CFD6
1B90FE30
```

```
  1D060355 1D0E0416 0414F3C2 1951DCA1 2729C0AA 52857ADC C4CFD61B
90FE300D
  06092A86 4886F70D 01010505 00038201 010070D3 EB58A268 9507168A
AFCFD256
  27118162 12376351 50E14BE4 7542F25E 8EABEFB7 9AF205C4 0CCB81E6
A3AFDEBA
  3C6225CB 90B37046 C7730C52 CE8E8AF4 C236EDDA 2C0545B8 EA627D81
BA1ACA07
  88DCDD54 A5D0B603 540382B3 06B962A2 1E95DA37 98AE6E25 B044F42A
F4E30566
  67A61A37 F1FE0EC8 AB2B2C7E 0BE20CEE B0BAF66F 558CD9E2 F186A11B
EA3EE95C
  9AC9614D D62FCA10 72EE1233 EE94225C B237ECFB D49C7833 6088D209
280D6C6E
  D0455572 FBDC56A6 0C4CCF0D FC66B4C9 56DBECB1 B2683FEF 8E8CFD33
BBFF4946
  99895C89 49E6A530 E47C4F2E 31900504 3AD77405 1C47CB13 7C3D37D9
E47E59D5
  40884BFC 1298DF18 278727CB EF5060E1 C3E5
        quit
crypto pki certificate chain TP-self-signed-4227613475

license boot level network-essentials addon dna-essentials

diagnostic bootup level minimal

spanning-tree mode rapid-pvst
spanning-tree extend system-id
memory free low-watermark processor 10308

redundancy
 mode sso

transceiver type all
 monitoring

class-map match-any system-cpp-police-ewlc-control
  description EWLC Control
class-map match-any system-cpp-police-topology-control
  description Topology control
class-map match-any system-cpp-police-sw-forward
  description Sw forwarding, L2 LVX data packets, LOGGING,
Transit Traffic
class-map match-any system-cpp-default
  description EWLC data, Inter FED Traffic
class-map match-any system-cpp-police-sys-data
```

```
  description Openflow, Exception, EGR Exception, NFL Sampled
Data, RPF Failed
class-map match-any system-cpp-police-punt-webauth
  description Punt Webauth
class-map match-any system-cpp-police-l2lvx-control
  description L2 LVX control packets
class-map match-any system-cpp-police-forus
  description Forus Address resolution and Forus traffic
class-map match-any system-cpp-police-multicast-end-station
  description MCAST END STATION
class-map match-any system-cpp-police-high-rate-app
  description High Rate Applications
class-map match-any system-cpp-police-multicast
  description MCAST Data
class-map match-any system-cpp-police-l2-control
  description L2 control
class-map match-any system-cpp-police-dot1x-auth
  description DOT1X Auth
class-map match-any system-cpp-police-data
  description ICMP redirect, ICMP_GEN and BROADCAST
class-map match-any system-cpp-police-stackwise-virt-control
  description Stackwise Virtual OOB
class-map match-any non-client-nrt-class
class-map match-any system-cpp-police-routing-control
  description Routing control and Low Latency
class-map match-any system-cpp-police-protocol-snooping
  description Protocol snooping
class-map match-any system-cpp-police-dhcp-snooping
  description DHCP snooping
class-map match-any system-cpp-police-ios-routing
  description L2 control, Topology control, Routing control, Low
Latency
class-map match-any system-cpp-police-system-critical
  description System Critical and Gold Pkt
class-map match-any system-cpp-police-ios-feature
  description
ICMPGEN,BROADCAST,ICMP,L2LVXCntrl,ProtoSnoop,PuntWebauth,MCASTDa
ta,Transit,DOT1XAuth,Swfwd,LOGGING,L2LVXData,ForusTraffic,ForusA
RP,McastEndStn,Openflow,Exception,EGRExcption,NflSampled,RpfFail
ed

policy-map system-cpp-policy

interface GigabitEthernet0/0
 vrf forwarding Mgmt-vrf
 no ip address
 shutdown
```

```
 negotiation auto

interface GigabitEthernet1/0/1
 no switchport
 ip address 192.168.6.2 255.255.255.0
 ipv6 address 2001:DB8:6::2/64
 ipv6 ospf 1 area 2

interface GigabitEthernet1/0/2

interface GigabitEthernet1/0/3

interface GigabitEthernet1/0/4

interface GigabitEthernet1/0/5

interface GigabitEthernet1/0/6

interface GigabitEthernet1/0/7

interface GigabitEthernet1/0/8

interface GigabitEthernet1/0/9

interface GigabitEthernet1/0/10

interface GigabitEthernet1/0/11

interface GigabitEthernet1/0/12

interface GigabitEthernet1/0/13

interface GigabitEthernet1/0/14

interface GigabitEthernet1/0/15

interface GigabitEthernet1/0/16

interface GigabitEthernet1/0/17

interface GigabitEthernet1/0/18

interface GigabitEthernet1/0/19

interface GigabitEthernet1/0/20

interface GigabitEthernet1/0/21
```

```
interface GigabitEthernet1/0/22

interface GigabitEthernet1/0/23

interface GigabitEthernet1/0/24

interface GigabitEthernet1/1/1

interface GigabitEthernet1/1/2

interface GigabitEthernet1/1/3

interface GigabitEthernet1/1/4

interface Vlan1
 ip address dhcp

router ospf 1
 router-id 7.7.7.7
 network 192.168.6.0 0.0.0.255 area 2

ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip http client source-interface Vlan1!


ipv6 router ospf 1
 router-id 7.7.7.7

control-plane
 service-policy input system-cpp-policy

line con 0
 exec-timeout 0 0
 stopbits 1
line aux 0
 stopbits 1
line vty 0 4
 login
line vty 5 15
 login

End
```

Commands confirm that multi-area OSPF has been set up correctly:

Ping and traceroute for IPv4 and IPv6:

```
MLS2#ping 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/2/4 ms
MLS2#traceroute 192.168.1.1
Type escape sequence to abort.
Tracing the route to 192.168.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.6.1 0 msec 0 msec 0 msec
  2 192.168.5.1 4 msec 4 msec 0 msec
  3 192.168.4.1 0 msec 0 msec 4 msec
  4 192.168.3.1 0 msec 0 msec 0 msec
  5 192.168.2.1 4 msec 4 msec 0 msec
  6 192.168.1.1 4 msec *  4 msec
MLS2#
```

```
MLS1#ping 192.168.6.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.6.2, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/1/4 ms
MLS1#traceroute 192.168.6.2
Type escape sequence to abort.
Tracing the route to 192.168.6.2
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.1.2 0 msec 4 msec 0 msec
  2 192.168.2.2 0 msec 4 msec 0 msec
  3 192.168.3.2 0 msec 0 msec 0 msec
  4 192.168.4.2 0 msec 0 msec 4 msec
  5 192.168.5.2 0 msec 0 msec 4 msec
  6 192.168.6.2 0 msec 4 msec *
MLS1#
```

```
MLS1#ping 2001:db8:acad:6::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:6::2, timeout is
2 seconds:
```

```
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/2/4 ms
MLS1#traceroute 2001:db8:acad:6::2
Type escape sequence to abort.
Tracing the route to 2001:DB8:ACAD:6::2

  1 2001:DB8:ACAD:1::2 4 msec 0 msec 0 msec
  2 2001:DB8:ACAD:2::2 4 msec 0 msec 0 msec
  3 2001:DB8:ACAD:3::2 0 msec 0 msec 0 msec
  4 2001:DB8:ACAD:4::2 0 msec 4 msec 0 msec
  5 2001:DB8:ACAD:5::2 4 msec 0 msec 4 msec
  6 2001:DB8:ACAD:6::2 0 msec 4 msec 0 msec
MLS1#


MLS2#ping 2001:db8:acad:1::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:1::1, timeout is
2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/2/4 ms
MLS2#traceroute 2001:db8:acad:1::1
Type escape sequence to abort.
Tracing the route to 2001:DB8:ACAD:1::1

  1 2001:DB8:ACAD:6::1 0 msec 0 msec 4 msec
  2 2001:DB8:ACAD:5::1 0 msec 0 msec 0 msec
  3 2001:DB8:ACAD:4::1 0 msec 0 msec 4 msec
  4 2001:DB8:ACAD:3::1 0 msec 0 msec 4 msec
  5 2001:DB8:ACAD:2::1 0 msec 0 msec 4 msec
  6 2001:DB8:ACAD:1::1 0 msec 0 msec 4 msec
MLS2#
```

This shows the packets being sent throughout the network

IPv6 and IPv4 Routes:

```
IPv6 Routing Table - default - 9 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static
route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX -
EIGRP external
```

```
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr
- Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 -
OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a -
Application
OI  2001:DB8:ACAD:1::/64 [110/3]
     via FE80::1, GigabitEthernet0/0/0
OI  2001:DB8:ACAD:2::/64 [110/2]
     via FE80::1, GigabitEthernet0/0/0
C   2001:DB8:ACAD:3::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:3::2/128 [0/0]
     via GigabitEthernet0/0/0, receive
C   2001:DB8:ACAD:4::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:4::1/128 [0/0]
     via GigabitEthernet0/0/1, receive
OI  2001:DB8:ACAD:5::/64 [110/2]
     via FE80::1, GigabitEthernet0/0/1
OI  2001:DB8:ACAD:6::/64 [110/3]
     via FE80::1, GigabitEthernet0/0/1
L   FF00::/8 [0/0]
     via Null0, receive
R3#

R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -
IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-
user static route
       o - ODR, P - periodic downloaded static route, H - NHRP,
l - LISP
       a - application route
       + - replicated route, % - next hop override, p -
overrides from PfR

Gateway of last resort is not set
```

```
O IA  192.168.1.0/24 [110/3] via 192.168.3.1, 00:06:14,
GigabitEthernet0/0/0
O IA  192.168.2.0/24 [110/2] via 192.168.3.1, 00:11:07,
GigabitEthernet0/0/0
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected,
GigabitEthernet0/0/0
L        192.168.3.2/32 is directly connected,
GigabitEthernet0/0/0
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.4.0/24 is directly connected,
GigabitEthernet0/0/1
L        192.168.4.1/32 is directly connected,
GigabitEthernet0/0/1
O IA  192.168.5.0/24 [110/2] via 192.168.4.2, 00:10:34,
GigabitEthernet0/0/1
O IA  192.168.6.0/24 [110/3] via 192.168.4.2, 00:10:21,
GigabitEthernet0/0/1
R3#
```

Another important command to show that multi-area OSPF is correctly configured is show ospf database (or show ospf data), when done on the multi-area routers it will be able show 2 areas at a time:

```
R2#show ospf database

          OSPFv3 1 address-family ipv6 (router-id 3.3.3.3)

                Router Link States (Area 0)

ADV Router       Age          Seq#         Fragment ID  Link count
Bits
 3.3.3.3         731          0x80000004   0            1
B
 4.4.4.4         703          0x80000007   0            2
None
 5.5.5.5         703          0x80000005   0            1
B

                Net Link States (Area 0)

ADV Router       Age          Seq#         Link ID    Rtr count
 4.4.4.4         739          0x80000001   6          2
 5.5.5.5         706          0x80000001   6          2

                Inter Area Prefix Link States (Area 0)
```

```
ADV Router      Age         Seq#        Prefix
 3.3.3.3        778         0x80000001  2001:DB8:ACAD:2::/64
 3.3.3.3        453         0x80000001  2001:DB8:ACAD:1::/64
 5.5.5.5        748         0x80000001  2001:DB8:ACAD:5::/64
 5.5.5.5        688         0x80000001  2001:DB8:ACAD:6::/64

                Link (Type-8) Link States (Area 0)

ADV Router      Age         Seq#        Link ID    Interface
 3.3.3.3        777         0x80000002  7          Gi0/0/1
 4.4.4.4        777         0x80000002  6          Gi0/0/1

                Intra Area Prefix Link States (Area 0)

ADV Router      Age         Seq#        Link ID    Ref-lstype
Ref-LSID
 4.4.4.4        739         0x80000001  6144       0x2002
6
 5.5.5.5        706         0x80000001  6144       0x2002
6

                Router Link States (Area 1)

ADV Router      Age         Seq#        Fragment ID  Link count
Bits
 1.1.1.1        382         0x80000003  0            1
None
 2.2.2.2        376         0x8000000B  0            2
None
 3.3.3.3        722         0x80000006  0            1
B

                Net Link States (Area 1)

ADV Router      Age         Seq#        Link ID    Rtr count
 2.2.2.2        382         0x80000001  6          2
 3.3.3.3        734         0x80000001  6          2

                Inter Area Prefix Link States (Area 1)

ADV Router      Age         Seq#        Prefix
 3.3.3.3        777         0x80000001  2001:DB8:ACAD:3::/64
 3.3.3.3        738         0x80000001  2001:DB8:ACAD:4::/64
 3.3.3.3        701         0x80000001  2001:DB8:ACAD:5::/64
 3.3.3.3        686         0x80000001  2001:DB8:ACAD:6::/64

                Link (Type-8) Link States (Area 1)
```

```
ADV Router       Age         Seq#        Link ID    Interface
 2.2.2.2         778         0x80000002  7          Gi0/0/0
 3.3.3.3         778         0x80000002  6          Gi0/0/0

                Intra Area Prefix Link States (Area 1)

ADV Router       Age         Seq#        Link ID    Ref-lstype
Ref-LSID
 2.2.2.2         382         0x80000001  6144       0x2002
6
 3.3.3.3         734         0x80000001  6144       0x2002
6
R2#


R2#show ospfv3 data

          OSPFv3 1 address-family ipv6 (router-id 3.3.3.3)

                Router Link States (Area 0)

ADV Router       Age         Seq#        Fragment ID  Link count
Bits
 3.3.3.3         769         0x80000004  0            1
B
 4.4.4.4         741         0x80000007  0            2
None
 5.5.5.5         741         0x80000005  0            1
B

                Net Link States (Area 0)

ADV Router       Age         Seq#        Link ID    Rtr count
 4.4.4.4         777         0x80000001  6          2
 5.5.5.5         744         0x80000001  6          2

                Inter Area Prefix Link States (Area 0)

ADV Router       Age         Seq#        Prefix
 3.3.3.3         816         0x80000001  2001:DB8:ACAD:2::/64
 3.3.3.3         491         0x80000001  2001:DB8:ACAD:1::/64
 5.5.5.5         786         0x80000001  2001:DB8:ACAD:5::/64
 5.5.5.5         726         0x80000001  2001:DB8:ACAD:6::/64

                Link (Type-8) Link States (Area 0)
```

```
ADV Router       Age         Seq#        Link ID    Interface
 3.3.3.3         815         0x80000002  7          Gi0/0/1
 4.4.4.4         815         0x80000002  6          Gi0/0/1

                Intra Area Prefix Link States (Area 0)

ADV Router       Age         Seq#        Link ID    Ref-lstype
Ref-LSID
 4.4.4.4         777         0x80000001  6144       0x2002
6
 5.5.5.5         744         0x80000001  6144       0x2002
6

                Router Link States (Area 1)

ADV Router       Age         Seq#        Fragment ID  Link count
Bits
 1.1.1.1         420         0x80000003  0            1
None
 2.2.2.2         414         0x8000000B  0            2
None
 3.3.3.3         760         0x80000006  0            1
B

                Net Link States (Area 1)

ADV Router       Age         Seq#        Link ID    Rtr count
 2.2.2.2         420         0x80000001  6          2
 3.3.3.3         772         0x80000001  6          2

                Inter Area Prefix Link States (Area 1)

ADV Router       Age         Seq#        Prefix
 3.3.3.3         815         0x80000001  2001:DB8:ACAD:3::/64
 3.3.3.3         776         0x80000001  2001:DB8:ACAD:4::/64
 3.3.3.3         739         0x80000001  2001:DB8:ACAD:5::/64
 3.3.3.3         724         0x80000001  2001:DB8:ACAD:6::/64

                Link (Type-8) Link States (Area 1)

ADV Router       Age         Seq#        Link ID    Interface
 2.2.2.2         816         0x80000002  7          Gi0/0/0
 3.3.3.3         816         0x80000002  6          Gi0/0/0

                Intra Area Prefix Link States (Area 1)
```

```
ADV Router       Age         Seq#        Link ID    Ref-lstype
Ref-LSID
 2.2.2.2         420         0x80000001  6144       0x2002
6
 3.3.3.3         772         0x80000001  6144       0x2002
6
R2#

ROUTER 4 DATABASE:
R4#show ospf database

          OSPFv3 1 address-family ipv6 (router-id 5.5.5.5)

                Router Link States (Area 0)

ADV Router       Age         Seq#        Fragment ID  Link count
Bits
 3.3.3.3         830         0x80000004  0            1
B
 4.4.4.4         800         0x80000007  0            2
None
 5.5.5.5         798         0x80000005  0            1
B

                Net Link States (Area 0)

ADV Router       Age         Seq#        Link ID    Rtr count
 4.4.4.4         835         0x80000001  6          2
 5.5.5.5         801         0x80000001  6          2

                Inter Area Prefix Link States (Area 0)

ADV Router       Age         Seq#        Prefix
 3.3.3.3         875         0x80000001  2001:DB8:ACAD:2::/64
 3.3.3.3         552         0x80000001  2001:DB8:ACAD:1::/64
 5.5.5.5         844         0x80000001  2001:DB8:ACAD:5::/64
 5.5.5.5         783         0x80000001  2001:DB8:ACAD:6::/64

                Link (Type-8) Link States (Area 0)

ADV Router       Age         Seq#        Link ID    Interface
 4.4.4.4         845         0x80000002  7          Gi0/0/0
 5.5.5.5         845         0x80000002  6          Gi0/0/0

                Intra Area Prefix Link States (Area 0)
```

```
ADV Router       Age         Seq#        Link ID    Ref-lstype
Ref-LSID
 4.4.4.4         835         0x80000001  6144       0x2002
6
 5.5.5.5         801         0x80000001  6144       0x2002
6

                Router Link States (Area 2)

ADV Router       Age         Seq#        Fragment ID  Link count
Bits
 5.5.5.5         775         0x80000004  0            1
B
 6.6.6.6         683         0x80000010  0            2
None
 7.7.7.7         683         0x80000003  0            1
None

                Net Link States (Area 2)

ADV Router       Age         Seq#        Link ID    Rtr count
 6.6.6.6         784         0x80000001  6          2
 6.6.6.6         683         0x80000001  7          2

                Inter Area Prefix Link States (Area 2)

ADV Router       Age         Seq#        Prefix
 5.5.5.5         845         0x80000001  2001:DB8:ACAD:4::/64
 5.5.5.5         801         0x80000001  2001:DB8:ACAD:3::/64
 5.5.5.5         801         0x80000001  2001:DB8:ACAD:2::/64
 5.5.5.5         550         0x80000001  2001:DB8:ACAD:1::/64

                Link (Type-8) Link States (Area 2)

ADV Router       Age         Seq#        Link ID    Interface
 5.5.5.5         844         0x80000002  7          Gi0/0/1
 6.6.6.6         824         0x80000001  6          Gi0/0/1

                Intra Area Prefix Link States (Area 2)

ADV Router       Age         Seq#        Link ID    Ref-lstype
Ref-LSID
 6.6.6.6         784         0x80000001  6144       0x2002
6
 6.6.6.6         683         0x80000001  7168       0x2002
7
```

```
R4#show ospfv3 database

          OSPFv3 1 address-family ipv6 (router-id 5.5.5.5)

                Router Link States (Area 0)

ADV Router       Age         Seq#         Fragment ID  Link count
Bits
 3.3.3.3         880         0x80000004  0            1
B
 4.4.4.4         850         0x80000007  0            2
None
 5.5.5.5         847         0x80000005  0            1
B

                Net Link States (Area 0)

ADV Router       Age         Seq#         Link ID    Rtr count
 4.4.4.4         885         0x80000001  6          2
 5.5.5.5         851         0x80000001  6          2

                Inter Area Prefix Link States (Area 0)

ADV Router       Age         Seq#         Prefix
 3.3.3.3         925         0x80000001  2001:DB8:ACAD:2::/64
 3.3.3.3         602         0x80000001  2001:DB8:ACAD:1::/64
 5.5.5.5         894         0x80000001  2001:DB8:ACAD:5::/64
 5.5.5.5         833         0x80000001  2001:DB8:ACAD:6::/64

                Link (Type-8) Link States (Area 0)

ADV Router       Age         Seq#         Link ID    Interface
 4.4.4.4         895         0x80000002  7          Gi0/0/0
 5.5.5.5         895         0x80000002  6          Gi0/0/0

                Intra Area Prefix Link States (Area 0)

ADV Router       Age         Seq#         Link ID    Ref-lstype
Ref-LSID
 4.4.4.4         885         0x80000001  6144       0x2002
6
 5.5.5.5         851         0x80000001  6144       0x2002
6

                Router Link States (Area 2)
```

```
ADV Router       Age         Seq#        Fragment ID  Link count
Bits
 5.5.5.5         824         0x80000004  0            1
B
 6.6.6.6         733         0x80000010  0            2
None
 7.7.7.7         733         0x80000003  0            1
None

                Net Link States (Area 2)

ADV Router       Age         Seq#        Link ID    Rtr count
 6.6.6.6         834         0x80000001  6          2
 6.6.6.6         733         0x80000001  7          2

                Inter Area Prefix Link States (Area 2)

ADV Router       Age         Seq#        Prefix
 5.5.5.5         894         0x80000001  2001:DB8:ACAD:4::/64
 5.5.5.5         850         0x80000001  2001:DB8:ACAD:3::/64
 5.5.5.5         850         0x80000001  2001:DB8:ACAD:2::/64
 5.5.5.5         600         0x80000001  2001:DB8:ACAD:1::/64

                Link (Type-8) Link States (Area 2)

ADV Router       Age         Seq#        Link ID    Interface
 5.5.5.5         894         0x80000002  7          Gi0/0/1
 6.6.6.6         874         0x80000001  6          Gi0/0/1

                Intra Area Prefix Link States (Area 2)

ADV Router       Age         Seq#        Link ID    Ref-lstype
Ref-LSID
 6.6.6.6         834         0x80000001  6144       0x2002
6
 6.6.6.6         733         0x80000001  7168       0x2002
7
```

Problem:

After configuring network, IPv4 worked as intended, while no IPv6 connectivity or neighbor relationships were established.

Cause of Problem:

In this lab the link-local address was manually configured to “fe80::1” on every interface, this caused the problem of duplicate address in the same network, causing errors in IPv6.

Solution:

To solve this issue, you can either manually set different link-local addresses (For example by alternating between fe80::1 and fe80::2 on each pair of interfaces in a network, you can use fe80::1 and fe80::2 again as long as it is in a different interface). Alternatively you can also let the router itself automatically assign the link-local address, the benefit manually configuring a link-local address is you can choose what it is set to rather than a automatically generated EUI-64 address is a long, complex string of hexadecimal character which could be hard to memorize.

Conclusion:

This lab demonstrated the configuration of a multi-area OSPF topology, knowing how to correctly set up multi-area OSPF is crucial, since it enhances network efficiency and stability. Multi-area OSPF limits the scope of updates throughout the network by dividing networks into smaller areas, localizing LSA flooding. Another important feature of multi-area OSPF is that it can reduce router load by allowing router summarization. Finally, multi-area OSPF also improves scalability of networks by creating hierarchical structures, letting networks grow reliably. All these reasons clearly show why configuring multi-area OSPF is a critical skill.